# Demand And Supply

Based on Ch. 3, Economics 10th ed., Parkin, Powell & Mathews

Course: Eco 101 (Introduction to Microeconomics)

# From Production to Markets

- Chapter 2 covered **production and trade**.
- After producing goods, **producers (sellers)** sell them to **buyers** in a **market**.
- **Market:** A place where buyers and sellers meet to trade.
  *Examples*: Brac cafeteria, bookstore, chaldal.com, bdjobs.com, Uber app.
- Markets are studied using the **Model of Demand and Supply** — one of the most important tools in economics.
- Chapter 3 introduces the **foundations** of this model.
- The first part of Chapter 3 focuses on the **buying (demand) side** of the market.

# WHAT IS DEMAND?

The word ***demand*** has a precise meaning in economics. If you demand something, the you

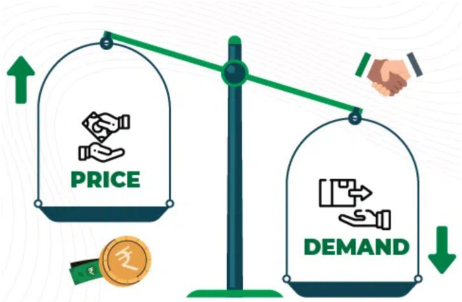
1. Want it,
2. Can afford it, and
3. Plan to buy it.

***The term demand refers to the entire relationship between the price of the good and the quantity demanded of the good.***

# The Law of Demand

As the price of a good rises, the quantity demanded of the good falls, and as the price of a good falls, the quantity demanded of the good rises, *ceteris paribus*.

$$P\uparrow Q_d\downarrow$$
$$P\downarrow Q_d\uparrow, \textit{ceteris paribus}$$




where, $P$=price and $Q_d$ = quantity demanded.

***Quantity demanded*** is the number of units of a good that individuals are willing and able to buy at a particular price during a particular period. For example, suppose individuals are willing and able to buy 100 TV dinners per week at a price of $4 per dinner. Therefore, 100 units is the quantity demanded of TV dinners at $4 per dinner.

# Why Does Quantity Demanded Go Down as Price Goes Up?

1. **Substitution Effect**

Although each good is unique, it has substitutes – other goods that can be used in its place. As the opportunity cost of a good rises, people buy less of that good and more of its substitutes.

2. **Income Effect**

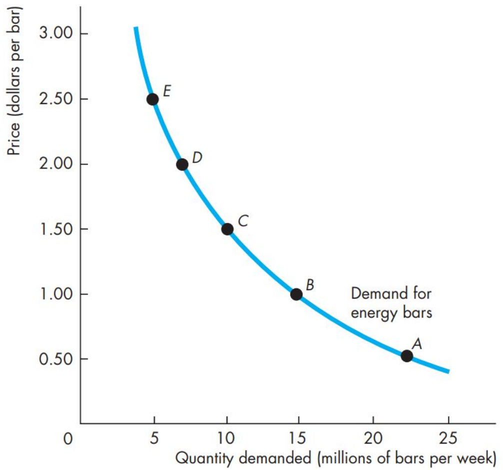
Faced with a higher price and unchanged income, people cannot afford to buy all the things they previously bought. They must decrease the quantities demanded of at least some goods and services, and normally, the good whose price has increased will be one of the goods that people buy less of.

# Demand Curve and Demand Schedule

A **demand curve** shows the relationship between the quantity demanded of a good and its price when all other influences on consumers' planned purchases remain the same. A **demand schedule** lists the quantities demanded at each price when all the other influences on consumers' planned purchases remain the same

| | Price (dollars per bar) | Quantity demanded (millions of bars per week) |
|---|---|---|
| *A* | 0.50 | 22 |
| *B* | 1.00 | 15 |
| *C* | 1.50 | 10 |
| *D* | 2.00 | 7 |
| *E* | 2.50 | 5 |

FIGURE 3.1 The Demand Curve




The **negative** relationship between price and $Q_d$ (quantity demanded) is called **<u>the law of demand</u>** and is represented by the **downward sloping** demand curve above.

# Continued

In the Model of Demand and Supply, we always put **price** *(the independent variable)* in the y-axis, and ***$Q_d$*** *(the dependent variable)* on the x-axis.

Therefore, when price (the independent variable of the graph) changes, $Q_d$ will change and there will a **movement** along the demand curve. This movement is referred to as a ***change in quantity demanded***.

However, when a factor other than price changes, there will be a **shift** of the demand curve. This shift is called a ***change in demand***.

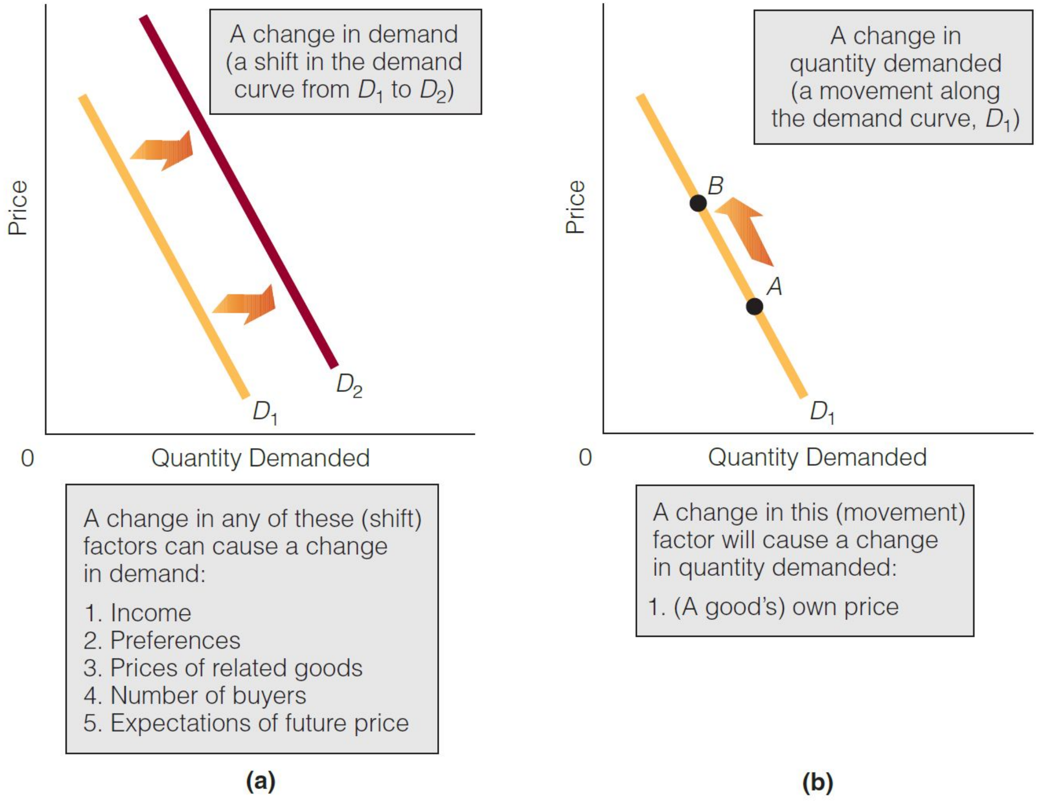
This is discussed in the following slides

# A *Change in Quantity Demanded* Versus a *Change in Demand*

**A Change in Demand Versus a Change in Quantity Demanded**

(a) A change in demand refers to a shift in the demand curve. A change in demand can be brought about by a number of factors. (see the exhibit and the text.)

(b) A change in quantity demanded refers to a movement along a given demand curve. A change in quantity demanded is brought about only by a change in (a good's) own price.



# Factors that Cause the Demand Curve to Shift

1) **Income:** As a person's income changes (increases or decreases), that individual's demand for a particular good may *rise, fall*, or *remain constant*.

- For a **normal good** (e.g., coffee), demand rises as income rises so the demand curve will shift right assuming ceteris paribus:

  *X* is a normal good: If income↑, then $D_X$ ↑, If income ↓, then $D_X$ ↓ .

- For an **inferior good** (e.g., local bus rides), demand falls as income rises so the demand curve will shift left assuming ceteris paribus:

  *Y* is an inferior good: If income ↑, then $D_Y$ ↓, If income ↓, then $D_Y$ ↑

# Continued

2) **Preferences:** People's preferences affect the amount of a good they are willing to buy at a particular price. Preferences are affected by information, advertisement, etc.

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- Change in preferences *in favor* of the good increases demand of the good, ceteris paribus
- Change in preferences *away from* the good decreases demand for the good, ceteris paribus

E.g., scientists have found out fish oil is good for our brains and hearts. This information may change the preference in favor of fish. Demand for fish will increase and the demand curve of fish in the fish market will shift right.

# Continued

3) **Population/Number of Buyers:**

- If number of buyers ↑ , then demand ↑ and the demand curve shifts right, ceteris paribus.
- The number of buyers may increase owing to a heightened birth rate, rising immigration, the migration of people from one region of the country to another, and so on.
- The number of buyers may decrease owing to an increased death rate, war, the migration of people from one region of the country to another, and so on.

4) **Expectations of Future Prices**:

- If buyers expect the price to ↑ in the future, then they are likely to purchase the product right now and therefore current demand ↑ and demand curve shifts right, ceteris paribus.
- If buyers expect the P to ↓ in the future, then buyers are likely to wait and buy the products in the future and thus current demand ↓ and demand curve shifts left, ceteris paribus.

*E.g., during Ramadan, price of food items usually increase, so buyers are likely to buy food items earlier and store them for consumption during Ramadan.*

# Continued

5) **Prices of Substitutes and Complements**: Most products have substitutes and complements.The demand of a product will be affected if price of substitutes or complements change.

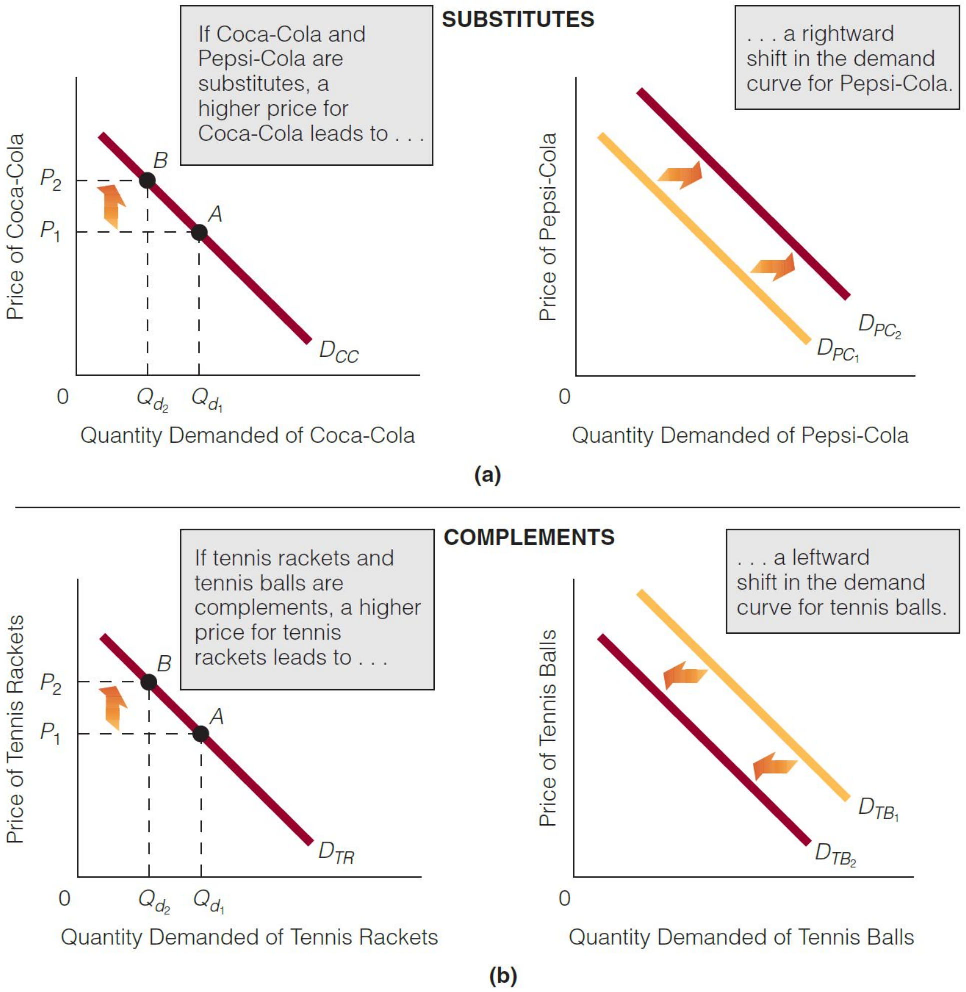
- **Substitute** goods satisfy similar needs and wants (Coffee and Tea; Coca-Cola and Pepsi-Cola).

- **Complements** are consumed together (Milk and Coffee; Pen and Paper; Tennis Rackets and Tennis Balls).

E.g., If price of tea ↑ , then some buyers will shift to coffee and demand for coffee ↑ and demand curve for coffee will shift right, ceteris paribus.

If price of milk ↑ , then buyers buy less milk and hence consumption of coffee decreases as well. Therefore, demand of coffee ↓ and demand curve will shift left, ceteris paribus.

## EXHIBIT 4




**Substitutes and Complements**

(a) Coca-Cola and Pepsi-Cola are substitutes: The price of one and the demand for the other are directly related. As the price of Coca-Cola rises, the demand for Pepsi-Cola increases. (b) Tennis rackets and tennis balls are complements: The price of one and the demand for the other are inversely related. As the price of tennis rackets rises, the demand for tennis balls decreases.

# WHAT IS SUPPLY?

If a firm supplies a good or a service, the firm

1. Has the resources and technology to produce it,
2. Can profit from producing it, and
3. Plans to produce it and sell it.

# The Law of Supply

The **law of supply states** that as the price of a good rises, the quantity supplied of the good rises, and as the price of a good falls, the quantity supplied of the good falls, *ceteris paribus*

$$P\uparrow Q_S\uparrow$$
$$P\downarrow Q_S\downarrow, \textit{ ceteris paribus}$$

where P=price and QS= quantity supplied

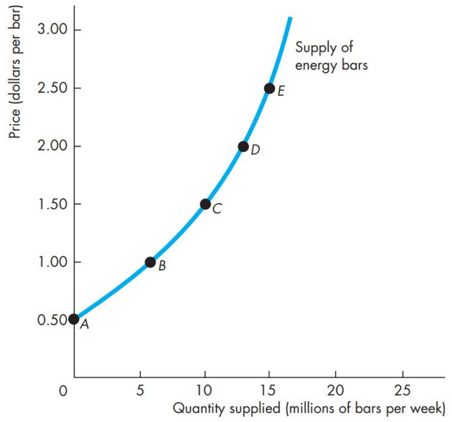
The **quantity supplied** of a good or service is the amount that producers plan to sell during a given time period at a particular price.

# Supply Curve and Supply Schedule

The term supply refers to the entire relationship between the quantity supplied and the price of a good.
Supply is illustrated by the supply curve and the supply schedule. The term quantity supplied refers to a point on a supply curve – the quantity supplied at a particular price.

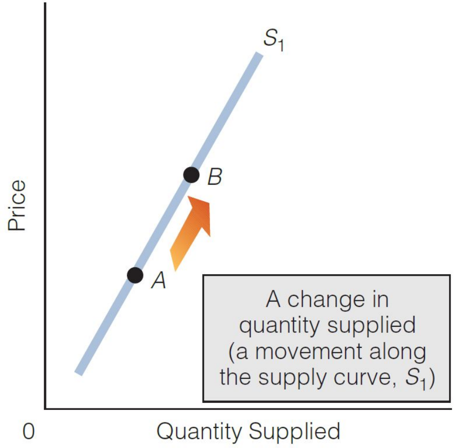
| | Price (dollars per bar) | Quantity supplied (millions of bars per week) |
|---|---|---|
| *A* | 0.50 | 0 |
| *B* | 1.00 | 6 |
| *C* | 1.50 | 10 |
| *D* | 2.00 | 13 |
| *E* | 2.50 | 15 |

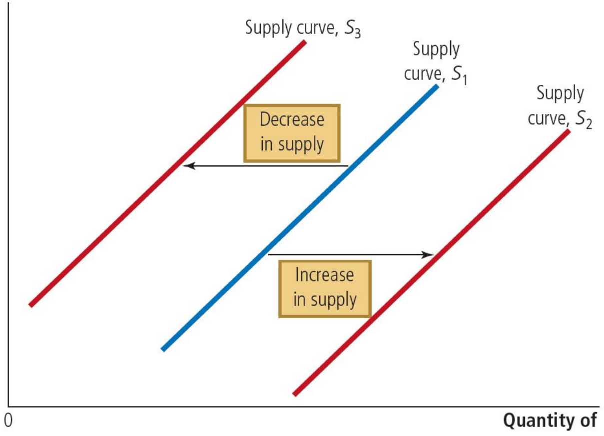
**FIGURE 3.4** The Supply Curve




The **positive** relationship between price and $Q_S$ (quantity supplied) is called **<u>the law of supply</u>** and is represented by the **upward sloping** supply curve above.

# A *Change in Quantity Supplied* Versus *a Change in Supply*




Changes in P (the independent variable in the graph) will cause a **movement** along the curve. This movement is referred to as a **change in *Qs*** (quantity supplied).

However, when a factor (outside of graph) other than P affects Qs, there will be a **shift** as shown in the graph. From *$S_1$ to $S_2$*, if there is an increase in supply, or from *$S_1$ to $S_3$*, if there is a decrease in supply. This shift is referred to as a **change in supply.**

# Factors that influence Change in Supply

A **shift in the supply curve** occurs when **any factor affecting selling plans changes — other than the price of the good itself**. There are **six key factors** that can lead to a change in supply:

1. **The prices of factors of production**
2. **The prices of related goods produced**
3. **Technology**
4. **The number of suppliers**
5. **Expected future prices**
6. **The state of nature**

# 1) Prices of Factors of Production:

- **Example:** To produce coffee, milk is a key input.
- If the **price of milk falls**, production becomes **cheaper**, assuming *ceteris paribus* (all else equal).
- As **costs decrease**, **profits rise**, encouraging producers to **increase output**.
- This leads to an **increase in the supply of coffee**, shifting the **supply curve to the right**.

# 2) The Prices of Relevant Goods Produced

### Substitutes in Production

- **Example:** Energy gel and energy bars.
- If the **price of energy gel rises**, producers may **shift resources** to gel and produce **fewer bars**.
- This causes the **supply of energy bars to decrease**.

### Complements in Production

- **Example:** Beef and cowhide.
- If the **price of beef rises**, more cattle are processed, which **increases the supply of cowhide** as a by-product

## 3) Technology:

- **Technological advancements** allow producers to **use fewer resources** to create the same output.
- This leads to **lower production costs**, **higher profits**, and an **increase in supply** (assuming *ceteris paribus*).
- As a result, the **supply curve shifts to the right**.
- **Example:** Improved farming techniques enable farmers to grow more crops with less labor. This reduces costs, boosts profit, and encourages more production — increasing supply.

## 4) Number of Sellers:

- When the **number of sellers increases**, the **market supply also increases** (assuming *ceteris paribus*).
- This causes the **supply curve to shift to the right**.
- **Example:** If more food stalls open in the BRAC cafeteria, the overall **availability of food increases**, raising the total supply.

## 5) Expectations of Future Prices:

- **Example:** If sellers **expect prices to rise in the future**, they may **hold back current supply** to sell later at a higher price.
- This causes a **decrease in current supply**, and the **supply curve shifts left**.

## 6) The State of Nature

- Refers to **natural conditions** affecting production — such as **weather and environmental events**.
- **Good weather** boosts agricultural supply, while **bad weather** (e.g., droughts, floods) reduces it.
- **Extreme events** like earthquakes, hurricanes, or wildfires can **disrupt production**, causing supply to fall.

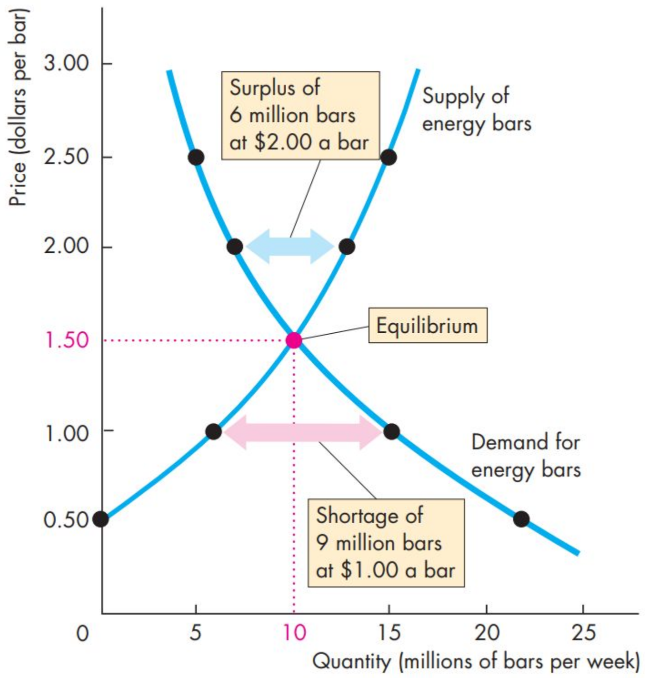
# Market Equilibrium

We have seen that when the price of a good rises, the quantity demanded *decreases* and the quantity supplied *increases.* We are now going to see how the price adjusts to coordinate buying plans and selling plans and achieve an equilibrium in the market.

An *equilibrium* is a situation in which opposing forces balance each other. Equilibrium in a market occurs when the price balances buying plans and selling plans. The **equilibrium price** is the price at which the quantity demanded equals the quantity supplied. The **equilibrium quantity** is the quantity bought and sold at the equilibrium price. A market moves toward its equilibrium because

- Price regulates buying and selling plans.
- Price adjusts when plans don't match.

**FIGURE 3.7** Equilibrium



# Market Equilibrium

- **Price as a Regulator**

The price of a good regulates the quantities demanded and supplied. If the price is too high, the quantity supplied exceeds the quantity demanded. If the price is too low, the quantity demanded exceeds the quantity supplied. There is one price at which the quantity demanded equals the quantity supplied.

- **Price Adjustments**

If the price is below equilibrium, there is a **shortage** and that if the price is above equilibrium, there is a **surplus.** But can we count on the price to change and eliminate a shortage or surplus? We can, because such price changes are beneficial to both buyers and sellers.

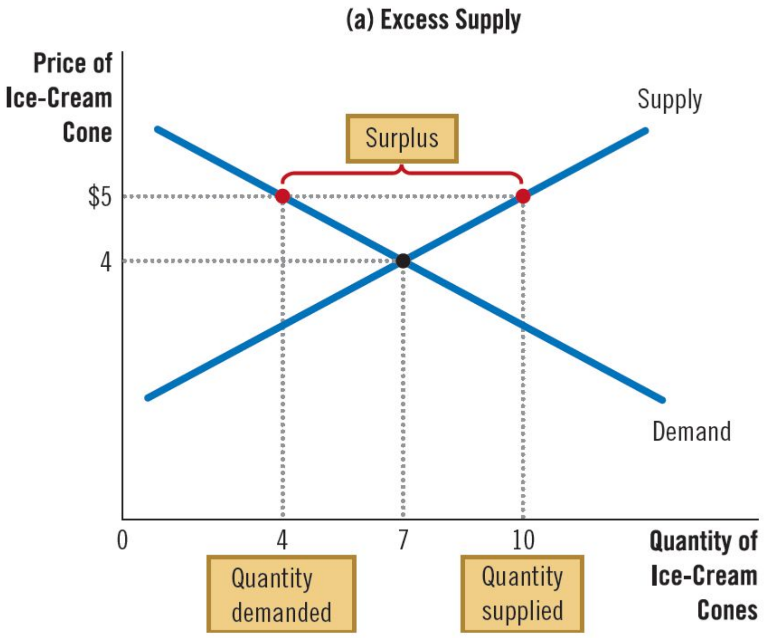
Why the price changes when there is a shortage or a surplus?

- A Shortage Forces the Price Up
- A Surplus Forces the Price Down

# Markets Not in Equilibrium

### Surplus of Ice Cream

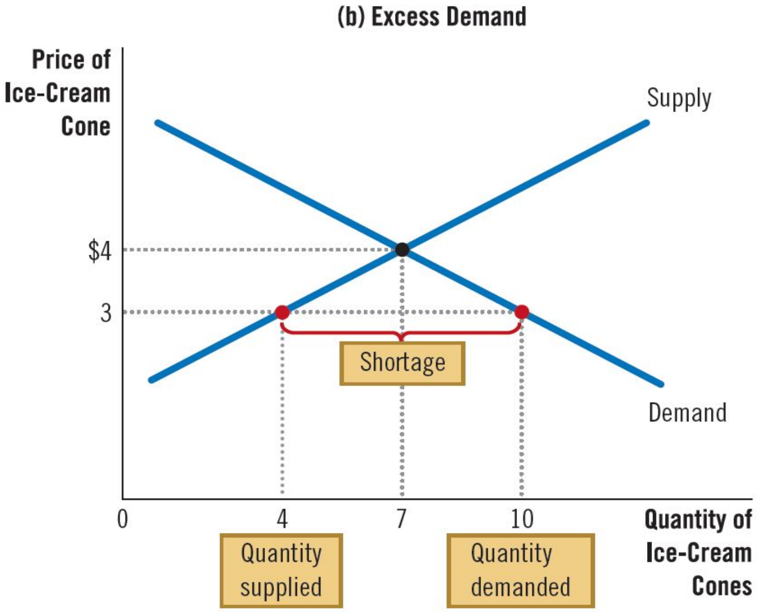
- Surplus: **Quantity supplied > Quantity demanded**
- Happens when **price is above equilibrium**
- E.g., at **$5**, supply = 10 cones, demand = 4 cones
- Creates **downward pressure on price**
- **Price falls** → demand rises, supply falls
- **Movement along** demand & supply curves
- Producers **cut price** to sell more
- Price keeps falling until **equilibrium** is reached



# Continued

## Shortage of Ice Cream

- Shortage: **Quantity demanded > Quantity supplied**
- Happens when **price is below equilibrium**
- E.g., at **$3**, demand = 10 cones, supply = 4 cones
- Creates **upward pressure on price**
- **Price rises** → demand falls, supply rises
- **Movement along** demand & supply curves
- Too many buyers, not enough goods → **producers raise price**
- Price rises until **equilibrium** is reached




***In both cases, the price adjustment moves the market toward the equilibrium of supply and demand.***

# Moving to Equilibrium

If a market is experiencing **surplus** ($Q_S > Q_D$), then suppliers will reduce the price to sell the excess goods. Price falls until it reaches the equilibrium price, where the market clears $Q_S = Q_D$.

If a market is experiencing **shortage** ($Q_D > Q_S$), then buyers bid the price up (they compete with each other for the limited goods, and sellers also raise the price). Price rises until the equilibrium price is reached, where the market clears $Q_D = Q_S$

Regardless of where the price starts, the market eventually reaches equilibrium. This phenomenon is called the **law of supply and demand**: The price of any good adjusts to bring the quantity supplied and the quantity demanded into balance. In most markets, surpluses and shortages are temporary.

# From One Equilibrium to Another

If the demand or supply curve shifts, then the market may move to a new equilibrium.

**Three steps to analyze changes in equilibrium:**

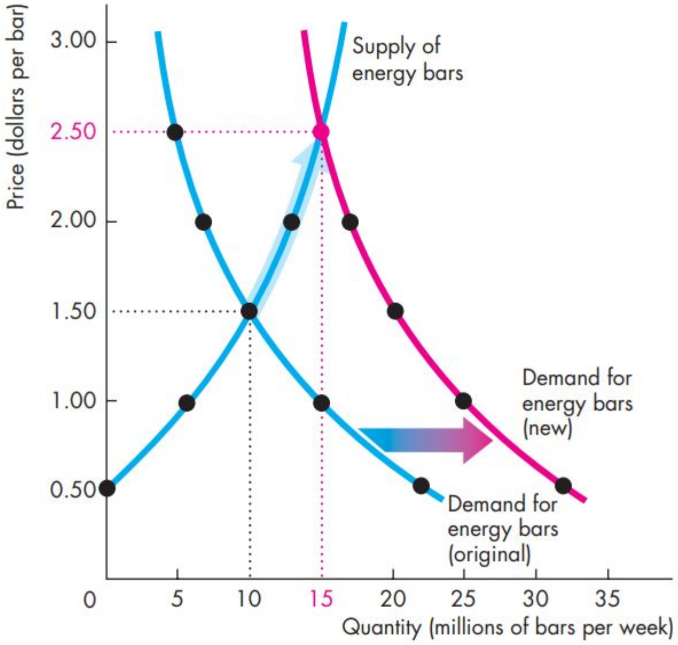
1. Decide whether the event shifts the supply curve, the demand curve, or, in some cases, both curves

1. Decide whether the curve shifts to the right or to the left

1. Use the supply-and-demand diagram
   - Compare the initial and the new equilibrium
   - See the effects on equilibrium price and quantity

# A Change in Market Equilibrium due to a Shift in Demand

FIGURE 3.8 The Effects of a Change in Demand



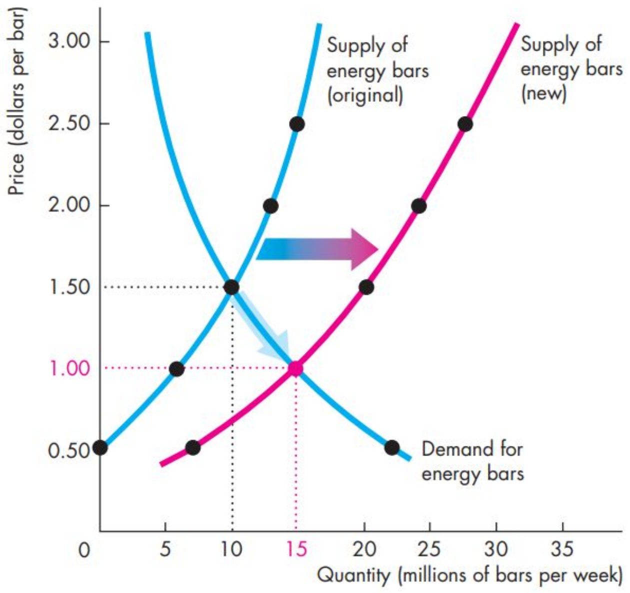

If more people join health clubs, the demand for energy bars increases.

- The demand curve shifts *right*
- The equilibrium price and quantity both rise
- Equilibrium price increases from \$1.50 to \$2.50
- Equilibrium quantity increases from 10 to 15 millions of bars.

**There is an increase in the quantity supplied but no change in supply—a movement along, but no shift of, the supply curve**.

# A Change in Market Equilibrium due to a Shift in Supply

FIGURE 3.9 The Effects of a Change in Supply

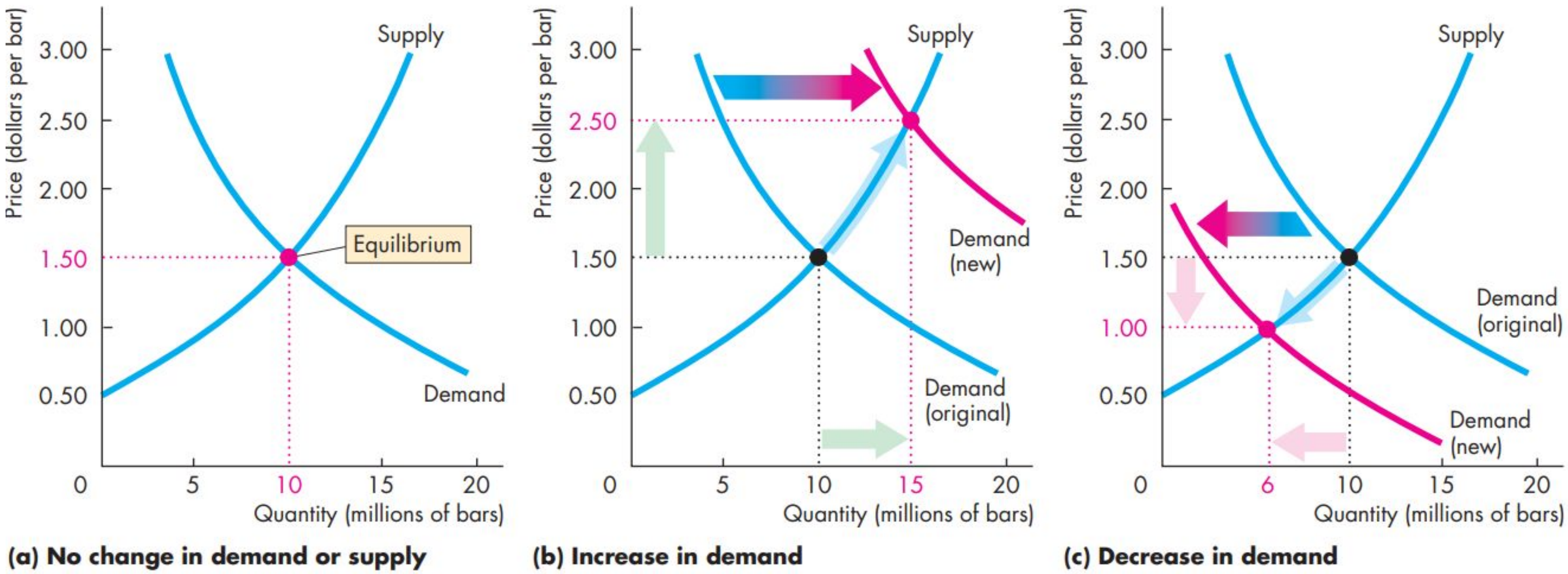



Due to new cost-saving technology, the supply of energy bar increases.

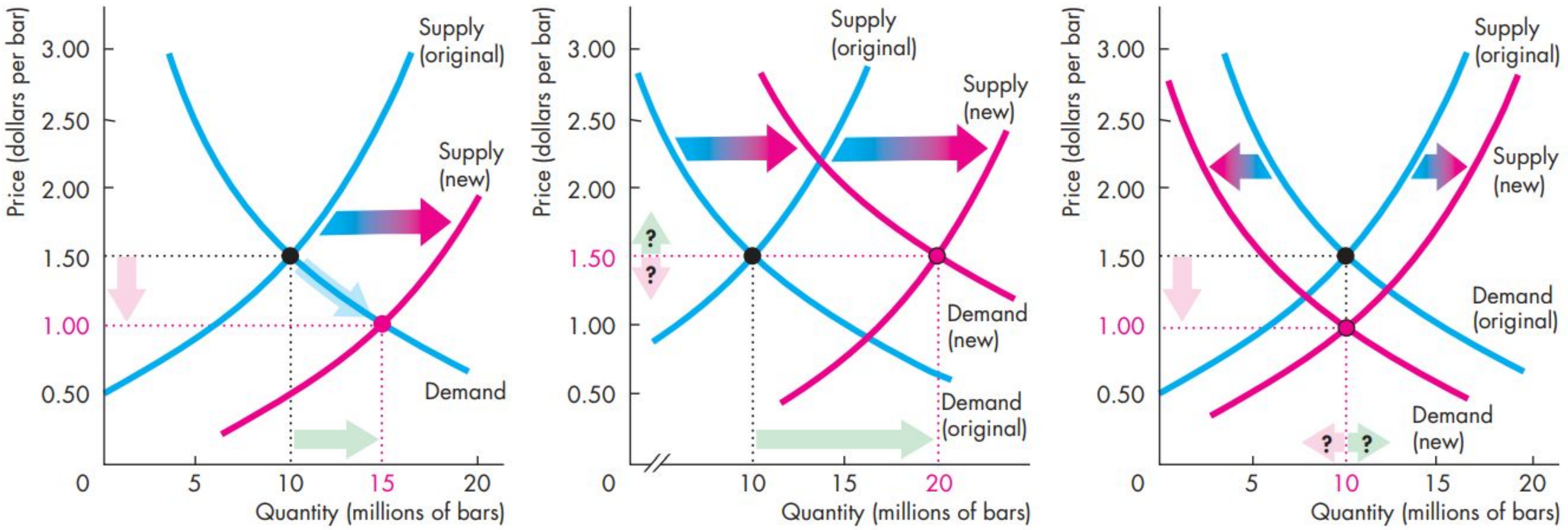
- The supply curve shifts *right*
- The equilibrium price falls, and the equilibrium quantity rises
- Equilibrium price of energy bars falls from $1.50 to $1.00
- Equilibrium quantity increases from 10 to 15

**There is an increase in the quantity demanded but no change in demand—a movement along, but no shift of, the demand curve.**

# All the Possible Changes in Demand and Supply

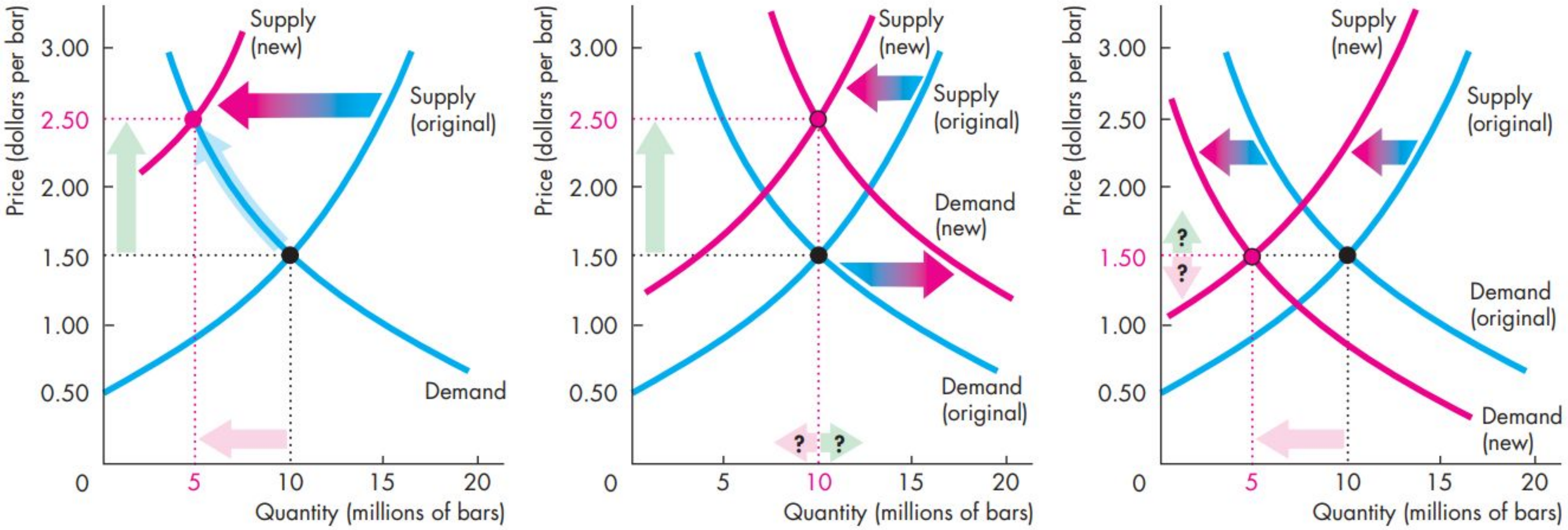


# All the Possible Changes in Demand and Supply




**(d) Increase in supply** **(e) Increase in both demand and supply** **(f) Decrease in demand; increase in supply**

# All the Possible Changes in Demand and Supply




**(g) Decrease in supply**

**(h) Increase in demand; decrease in supply**

**(i) Decrease in both demand and supply**